



(12) **United States Patent**
Baik et al.

(10) **Patent No.:** **US 12,416,684 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEM FOR DIAGNOSING BATTERY CELL USING SUBSTRATE MANUFACTURED BY LDCMOS PROCESS**

(71) Applicants: **AIRPOINT CO., LTD., Cell Point Co., LTD., Daejeon (KR)**

(72) Inventors: **Sung Jun Baik, Daejeon (KR); Jun Ho Cho, Daejeon (KR)**

(73) Assignees: **AIRPOINT CO., LTD., Daejeon (KR); Cell Point Co., LTD., Daejeon (KR)**

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 160 days.

(21) Appl. No.: **18/332,857**

(22) Filed: **Jun. 12, 2023**

(65) **Prior Publication Data**
US 2024/0337703 A1 Oct. 10, 2024

(30) **Foreign Application Priority Data**
Apr. 6, 2023 (KR) 10-2023-0045085

(51) **Int. Cl.**
G01R 31/396 (2019.01)
G01R 31/371 (2019.01)
G01R 31/3842 (2019.01)

(52) **U.S. Cl.**
CPC **G01R 31/396** (2019.01); **G01R 31/371** (2019.01); **G01R 31/3842** (2019.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2018/0233930 A1* 8/2018 Chiyajo H02J 7/0014
2019/0361075 A1* 11/2019 Lee H01M 10/4207

FOREIGN PATENT DOCUMENTS

CN	113690958	11/2021
KR	10-2019-0092088	8/2019
KR	10-2020-0119572	10/2020
KR	10-2022-0129735	9/2022

OTHER PUBLICATIONS

Office Action for Korea Patent Application No. 10-2023-0045085, mailed Nov. 22, 2024.

* cited by examiner

Primary Examiner — Jas A Sanghera

(57) **ABSTRACT**

A system for diagnosing a battery cell using a substrate manufactured by an LDCMOS process, includes a master BMS and a plurality of direct BMSs, wherein a battery cell connection unit included in each direct BMS makes direct contact with a positive electrode terminal and a negative electrode terminal of each of a plurality of battery cells of a battery module to measure raw data of voltages of the plurality of battery cells and raw data of a current of the battery module, and in order to accurately process raw data of the voltages of the plurality of battery cells and raw data of the currents, a first substrate unit manufactured by the LDCMOS process and a second substrate unit including a configuration to wirelessly transmit and receive a signal generated and processed by the first substrate unit to and from the master BMS are provided.

9 Claims, 9 Drawing Sheets

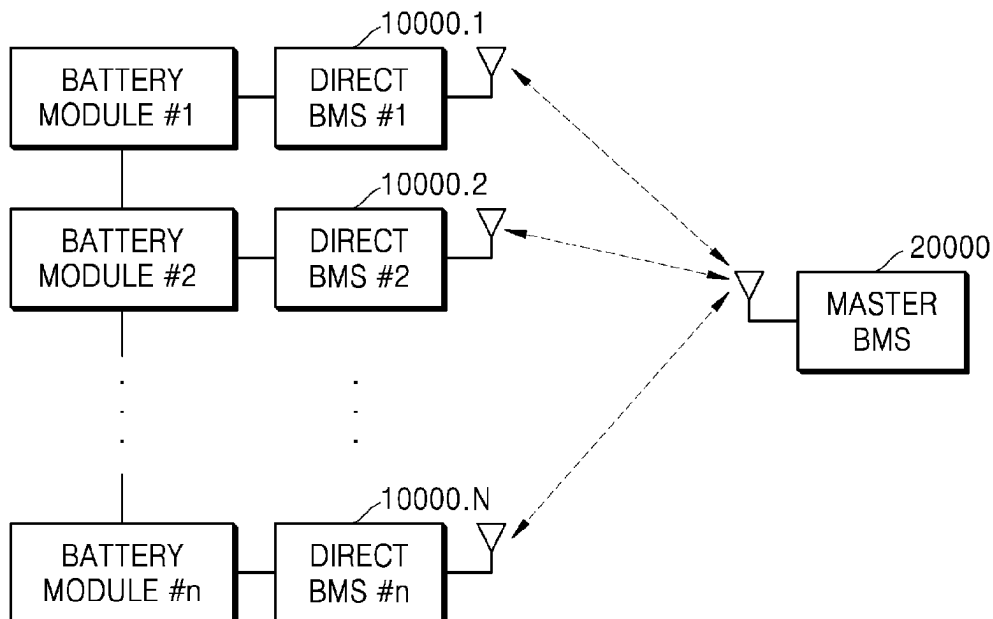


FIG. 1

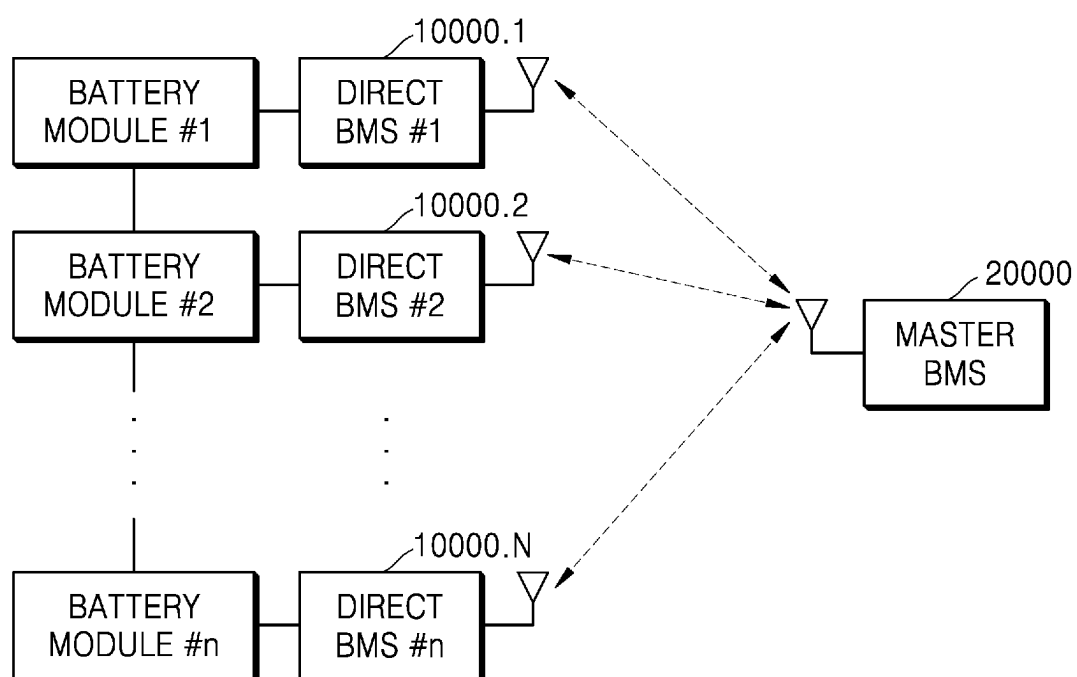


FIG. 2

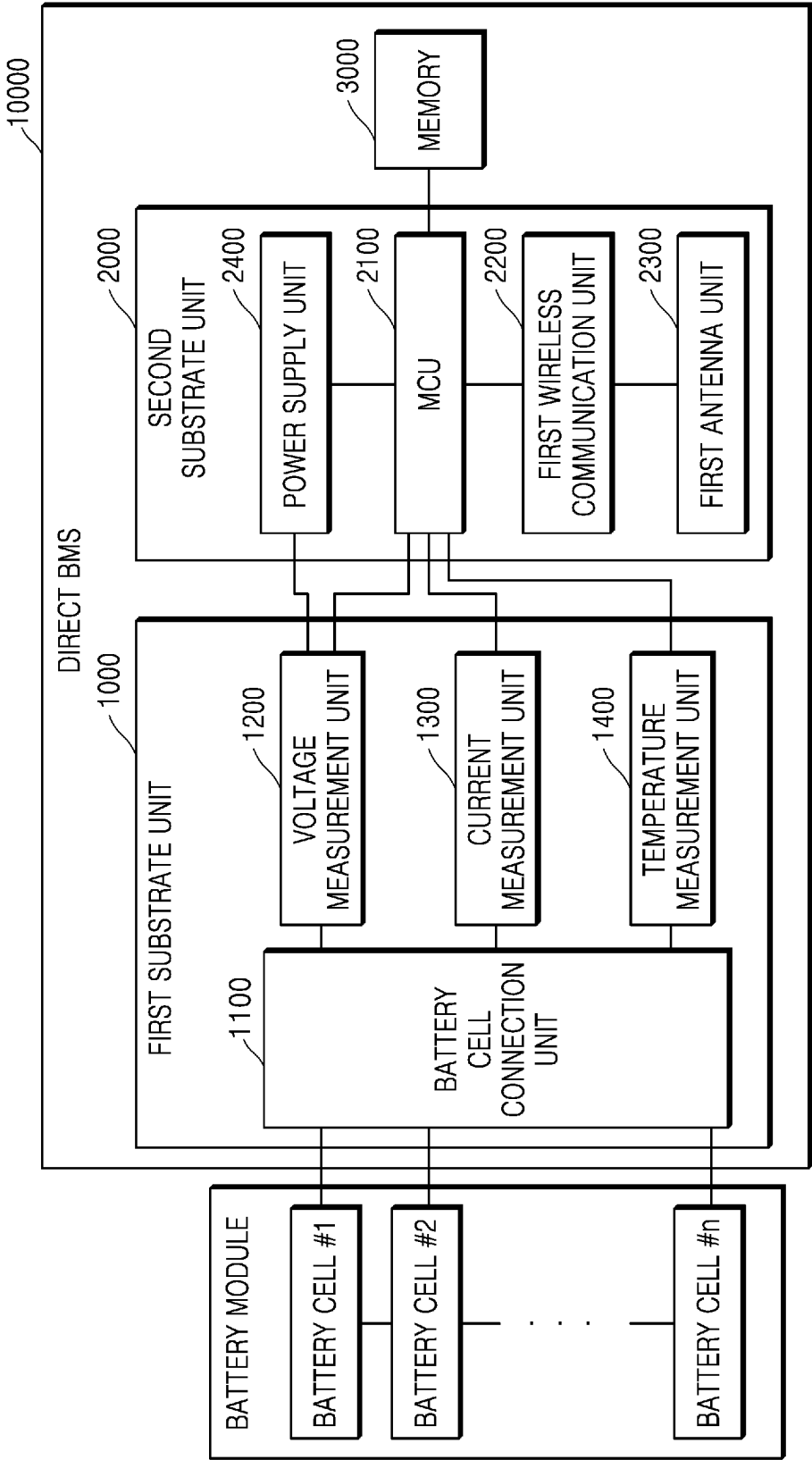


FIG. 3

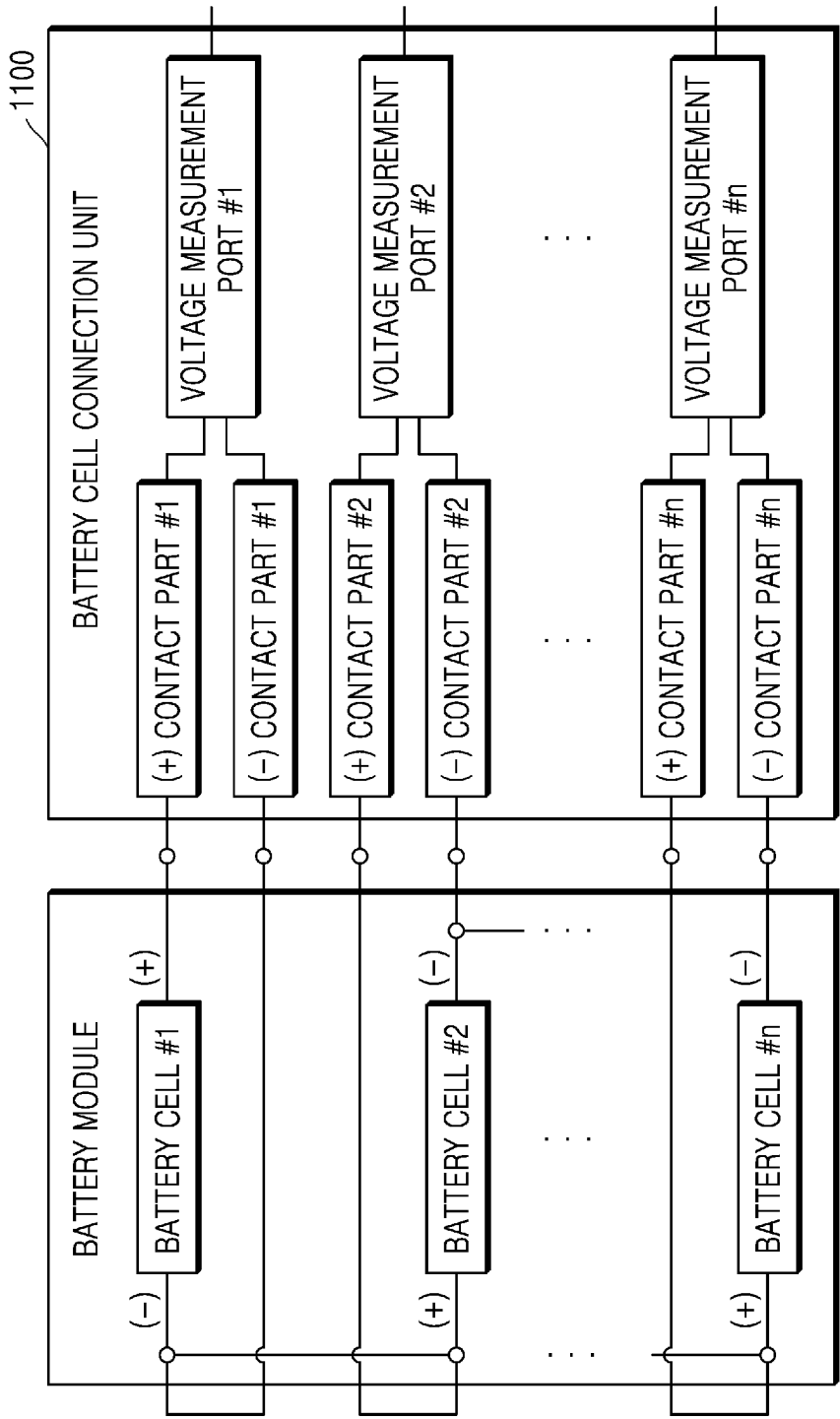


FIG. 4

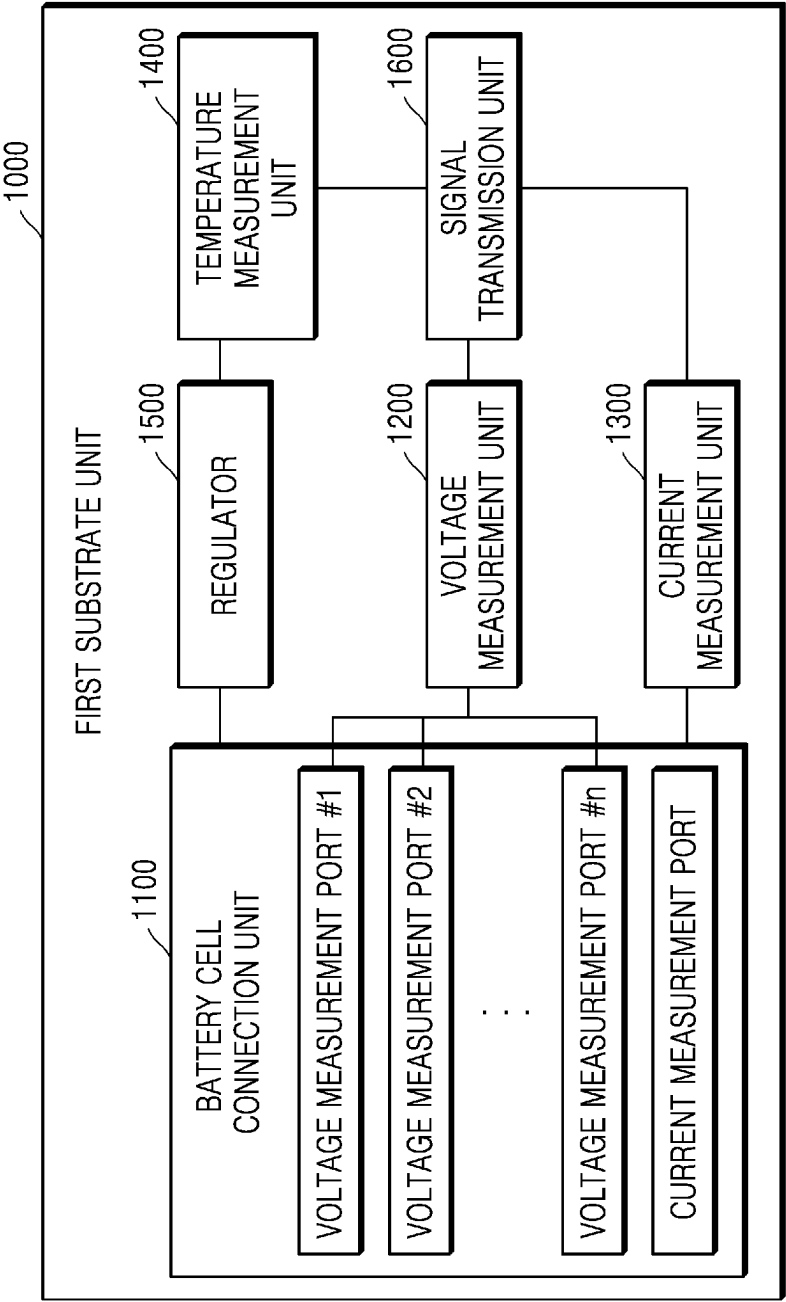


FIG. 5

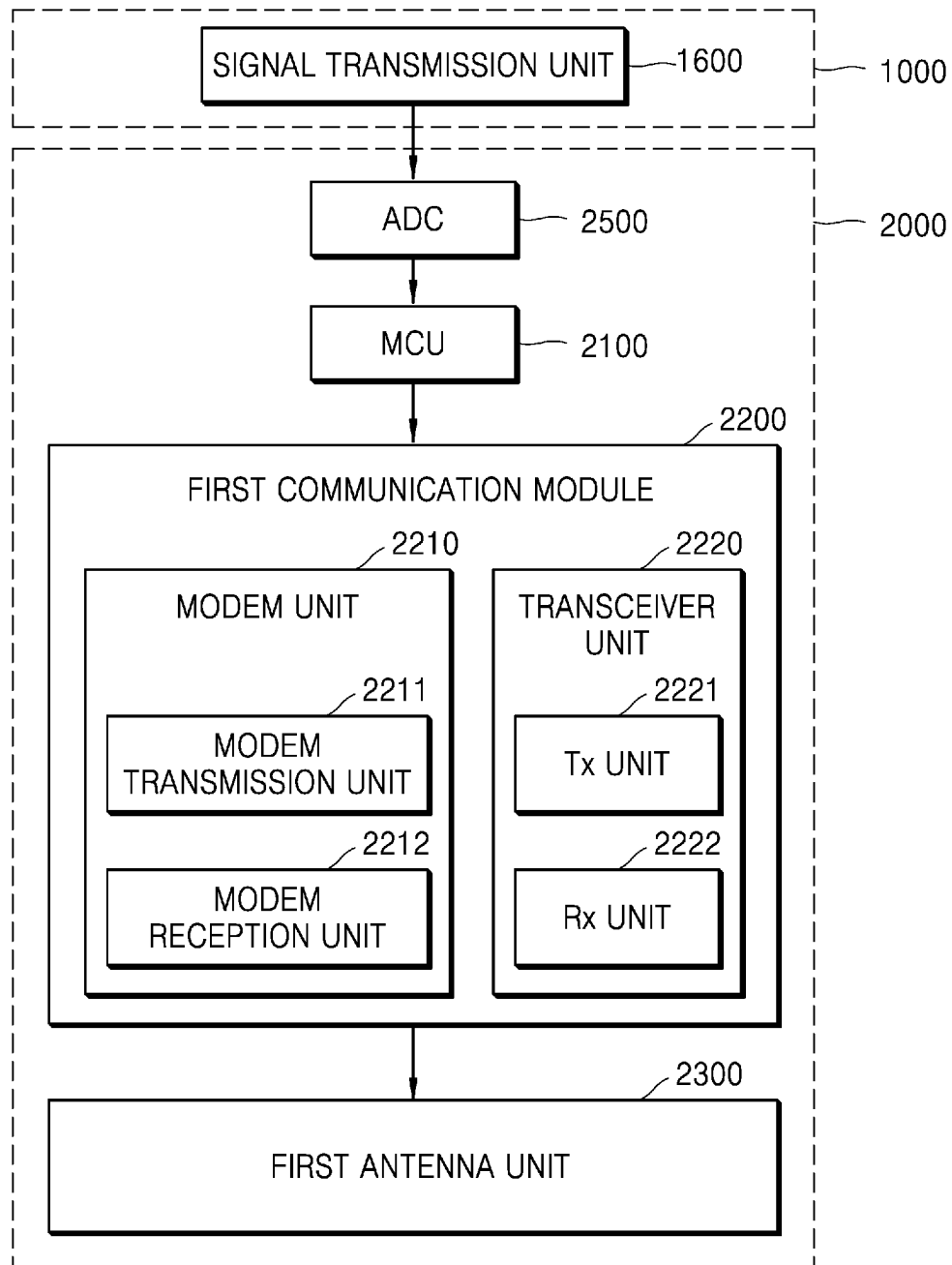


FIG. 6

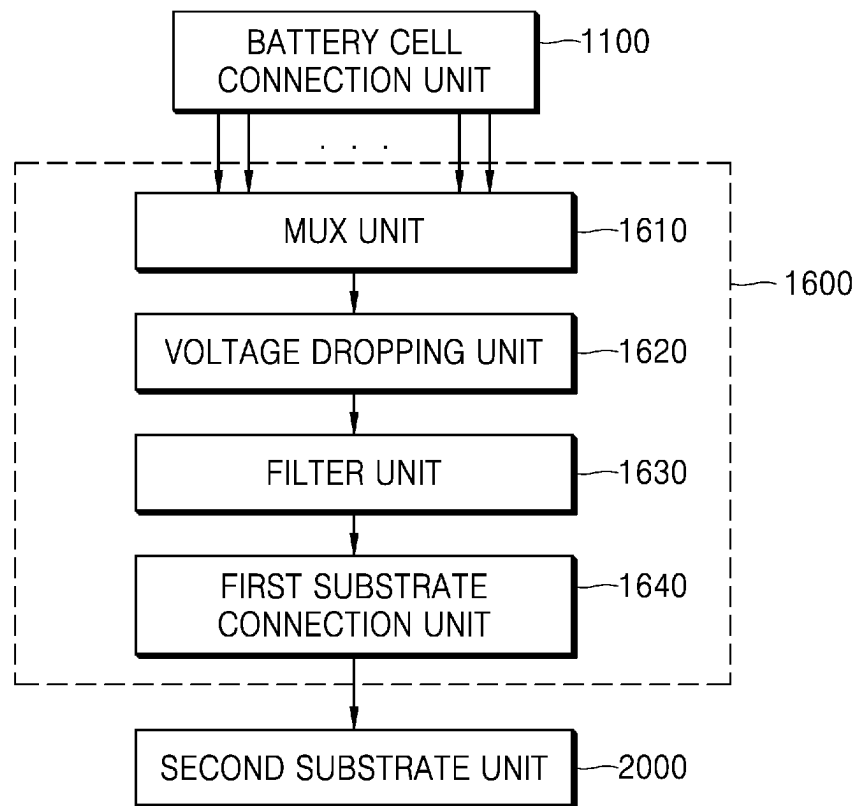


FIG. 7

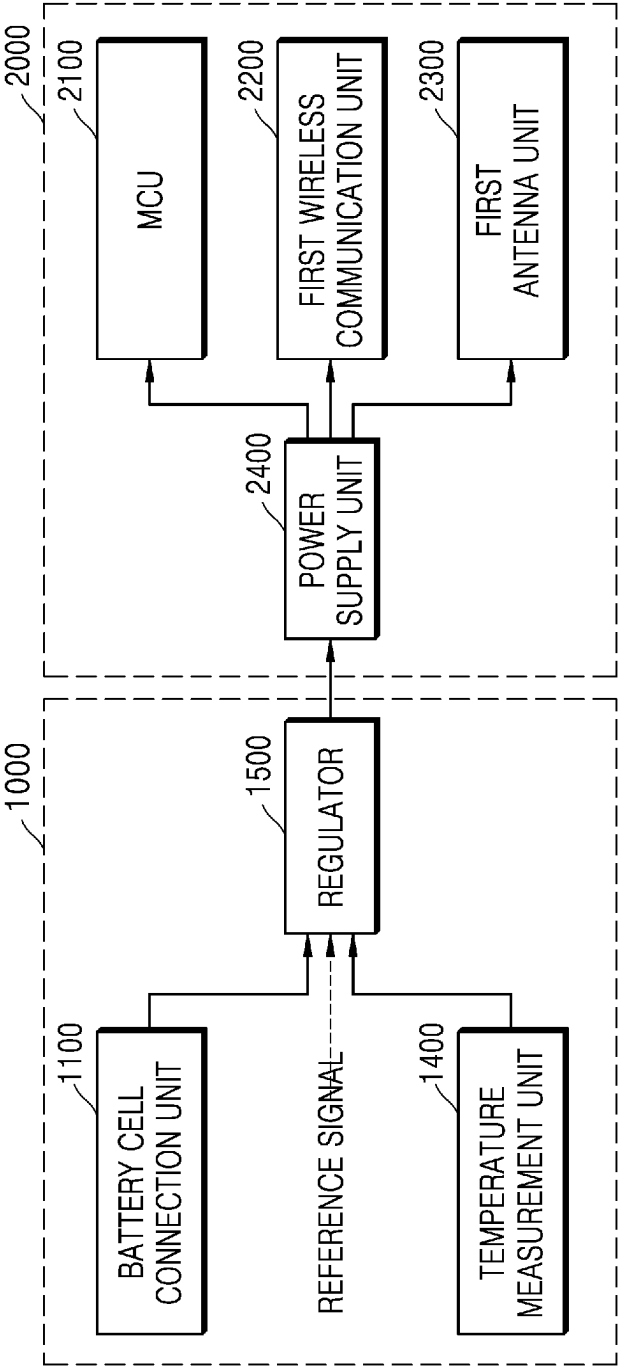


FIG. 8

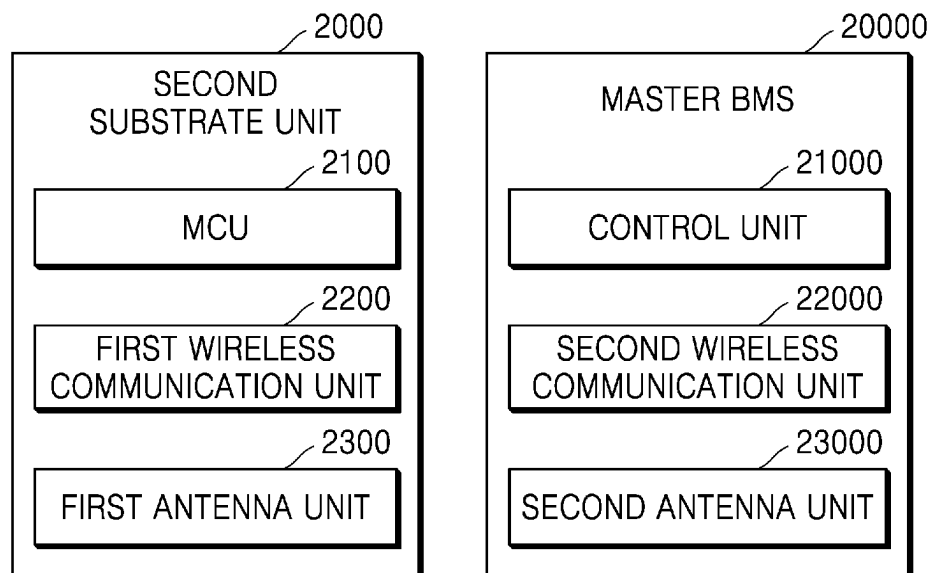
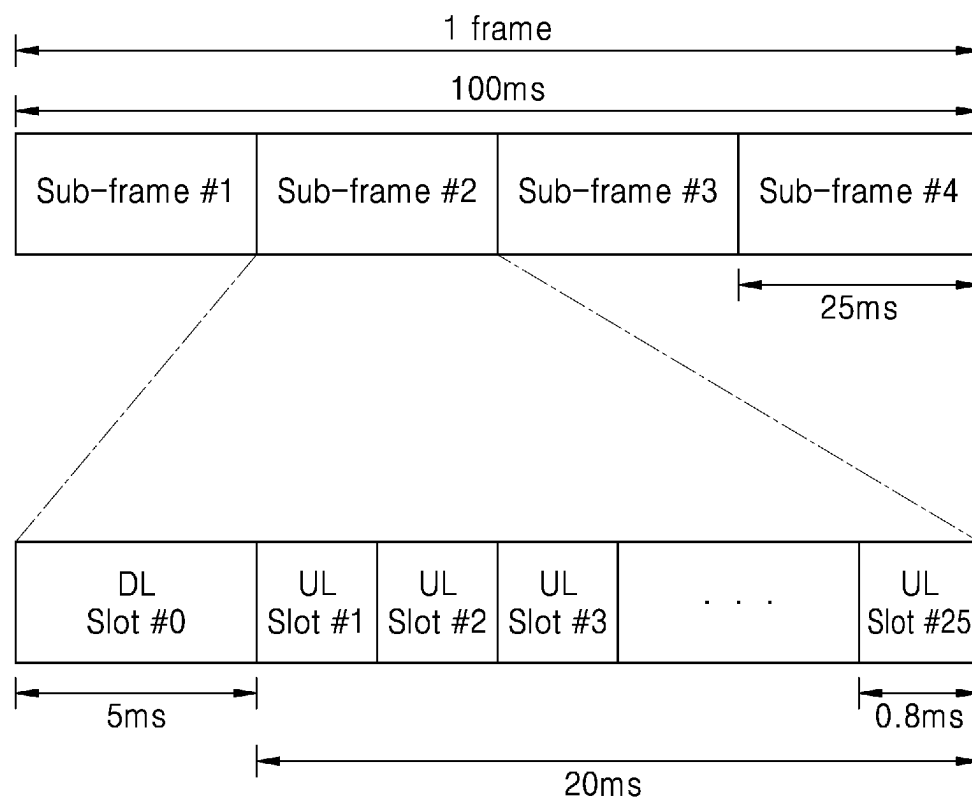


FIG. 9



1

SYSTEM FOR DIAGNOSING BATTERY CELL USING SUBSTRATE MANUFACTURED BY LDCMOS PROCESS

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a system for diagnosing a battery cell using a substrate manufactured by an LDCMOS process, and more specifically, to a system for diagnosing a battery cell using a substrate manufactured by an LDCMOS process, in which the system includes a master BMS and a plurality of direct BMSs, wherein a battery cell connection unit included in each of the direct BMSs comes into direct contact with a positive electrode terminal and a negative electrode terminal of each of a plurality of battery cells included in a battery module to measure raw data of a voltage of each of the battery cells and raw data of a current of the battery module, and in order to accurately process the raw data of the current of the plurality of battery cells and the raw data of the current, a first substrate unit manufactured by the LDCMOS process and a second substrate unit including a configuration to wirelessly transmit and receive a signal generated and processed by the first substrate unit to and from the master BMS are provided.

2. Description of the Related Art

Recently, environmental regulations for vehicles having internal combustion engines have been gradually severe all over the world, and accordingly, demands for vehicles, such as electric vehicles (EVs) or hybrid vehicles (MHEVs, PHEVs and FHEVs), using battery power as motive power have increased. As a result, automotive companies around the world have been increasingly interested in battery management technologies for improving efficiency, performance and durability of batteries, and researches and developments related to the technologies are being actively conducted in various companies and research organizations.

Particularly, a voltage, a current, and a temperature are the most important indicators among various indicators capable of determining a condition of a battery installed in a vehicle. A battery management system (BMS) for managing and controlling the battery can determine the state of the battery by measuring the voltage, current, temperature and the like of the battery installed in the vehicle. In general, the BMS includes: a plurality of slave BMS for managing at least one battery module; and a master BMS for controlling the slave BMSs.

Meanwhile, when forming a semiconductor chip or module accommodated in a conventional BMS, since the semiconductor chip or module is connected to a plurality of battery cells, the semiconductor chip or module is manufactured by a bipolar-CMOS-DMOS (BCD) process such that the semiconductor chip or module is operable even at a high voltage of several tens to several hundreds of volts (V). The BCD process is widely used as a power semiconductor process because it can withstand a high voltage by implementing Bipolar, CMOS, and DMOS in one chip. Examples of battery-related technologies using the BCD process as described above include a gate driving circuit for a cell balancing circuit capable of withstanding a high voltage, as disclosed in Korean Unexamined Patent Publication No. 10-2020-0119572.

However, as described above, when a state of the battery is monitored through the BCD process, it is difficult to

2

monitor each of battery the cells due to process characteristics, and the accuracy of sensing monitoring information about each of the battery cells is significantly reduced. Therefore, as an alternative to the technology using the BCD process, a BMS including a chip or module manufactured by an LDCMOS process has been suggested. However, in connecting and operating a master BMS and a plurality of direct BMSs, there is no BMS-related technology using a chip or module manufactured by the LDCMOS process without any type of relay BMS that relays or collects information about a plurality of battery cells between the master BMS and the direct BMS and transmits the collected information.

RELATED ART DOCUMENT

Patent Document

(Patent Document 0001) Korean Unexamined Patent Publication No. 10-2020-0119572 (published on Oct. 20, 2020)

SUMMARY OF THE INVENTION

An object of the present invention is to provide a system for diagnosing a battery cell using a substrate manufactured by an LDCMOS process, and more specifically, to a system for diagnosing a battery cell using a substrate manufactured by an LDCMOS process, in which the system includes a master BMS and a plurality of direct BMSs, wherein a battery cell connection unit included in each of the direct BMSs comes into direct contact with a positive electrode terminal and a negative electrode terminal of each of a plurality of battery cells included in a battery module to measure raw data of a voltage of each of the battery cells and raw data of a current of the battery module, and in order to accurately process the raw data of the current of the plurality of battery cells and the raw data of the current, a first substrate unit manufactured by the LDCMOS process and a second substrate unit including a configuration to wirelessly transmit and receive a signal generated and processed by the first substrate unit to and from the master BMS are provided.

To solve the above problem, one embodiment of the present invention provides a system for diagnosing a battery cell located inside a vehicle, in which the system a plurality of direct BMSs located inside the vehicle and electrically connected to a plurality of battery cells included in a battery module; and a master BMS located inside the vehicle and configured to wirelessly communicate with the direct BMSs, wherein the direct BMS includes: a first substrate unit operable in a first power range and including a substrate and an element; and a second substrate unit operable in a second power range lower than the first power range and including a substrate and an element, the first substrate unit includes: a battery cell connection unit including ports that physically and electrically come into direct contact with an exposed electrode of each of the plurality of battery cells; a voltage measurement unit electrically connected to the battery cell connection unit and configured in a form of a circuit to measure a voltage of each of the battery cells; and a current measurement unit electrically connected to the battery cell connection unit and configured in a form of a circuit to measure a current of the battery cell or the battery module, and the second substrate unit includes: an MCU; a first wireless communication unit connected to the MCU to wirelessly communicate with the master BMS; and a power supply unit configured to supply power to an element inside

the second substrate unit by receiving power from the first substrate unit while lowering a voltage of power of the plurality of battery cells.

In one embodiment of the present invention, both a positive electrode terminal and a negative electrode terminal of each of the plurality of battery cells may come into direct contact with the ports included in the battery cell connection unit, and the voltage measurement unit may be connected to each port to measure a voltage for each of the battery cells.

In one embodiment of the present invention, the first substrate unit may include a signal transmission unit configured to receive measurement information including the current and the voltage of the plurality of battery cells measured by the current measurement unit and the voltage measurement unit, respectively, to output the measurement information to the second substrate unit as an analog electrical signal, the second substrate unit may include an ADC, and the ADC may convert the analog electrical signal received from the signal transmission unit into a digital electrical signal to transmit the digital electrical signal to the MCU.

In one embodiment of the present invention, the signal transmission unit include: a multiplexer (MUX) unit including a circuit configured to output one of electrical signals for each of the plurality of battery cells under a control of the MCU; a voltage dropping unit including a circuit configured to drop a voltage signal received from the voltage measurement unit under the control of the MCU; and a filter unit including a circuit configured to reduce noise of the electrical signal received from the voltage measurement unit, and the signal transmission unit may transmit an analog voltage signal to the ADC after reducing a magnitude of the voltage signal, which is transmitted through the battery cell connection unit, to a predetermined reference or less.

In one embodiment of the present invention, the first substrate unit may be manufactured by an LDCMOS process.

In one embodiment of the present invention, the first substrate unit may operate in the first power range of 32 V to 36 V, and the second substrate unit may operate in the second range of 5 V or less.

In one embodiment of the present invention, the first substrate unit may include a regulator configured to supply power to the power supply unit by receiving the power of all of the plurality of battery cells, and the regulator may supply the power to the power supply unit by reducing a magnitude of a voltage signal of all of the plurality of battery cells to a predetermined reference or less when a reference signal is received and temperature information received from a temperature measurement unit is lower than a predetermined reference, and stop power supply to the power supply unit when the reference signal is not received or the temperature information received from the temperature measurement unit is equal to or higher than the predetermined reference.

In one embodiment of the present invention, the second substrate unit may include: a modem unit including a modem transmission unit configured to perform modulation for transmitting information received from the MCU to the master BMS, and a modem reception unit configured to perform demodulation for transmitting the signal received from the master BMS to the MCU; and a first wireless communication unit including a configuration to broadcast the signal modulated by the modem transmission unit through a first antenna unit; and an Rx unit including a configuration to process the signal received through the first antenna unit.

In one embodiment of the present invention, a communication frequency channel for communication between the direct BMS and the master BMS may include a plurality of narrow band channels divided at intervals of 200 KHz in a 900 MHz band.

According to one embodiment of the present invention, the system is designed based on of individual cells and power to be supplied when a state of the battery is sensed, and the system may thus be operated at a relatively low voltage, thereby using a substrate implemented by the LDCMOS process rather than the conventional BCD process, and accordingly, it is possible to improve sensing accuracy for a battery.

According to one embodiment of the present invention, a configuration is adopted in which the first substrate unit which may measure the voltage, current, and temperature for the plurality of battery cells and is implemented by the LDCMOS process, and the second substrate unit which may wirelessly transmit information measured by the first substrate unit to the master BMS are implemented by different processes, and the first substrate unit and the second substrate unit are combined into one system semiconductor through an MCP method, so that it is possible to reduce manufacturing costs.

According to one embodiment of the present invention, the ADC is included in the second substrate unit, not the first substrate unit implemented by the LDCMOS process, so that it is possible to maximize performance thereof while minimizing the size of the ADC.

According to one embodiment of the present invention, the direct BMS is configured to physically and electrically come into direct contact with the battery cells, thereby measuring the voltage and the current for the battery cells, so that noise generated due to channel switching can be prevented in the related art in which the battery and the direct BMS are indirectly connected through a cable or the like, and accordingly, the reliability of data measured for the battery cells can be increased.

According to one embodiment of the present invention, a master BMS and a plurality of direct BMSs connected to each of a plurality of battery modules perform asynchronous wireless communication, so that the problems caused by applying or installing an isolator to a plurality of positions to prevent electro-magnetic interference from being induced in the wired communication system of the related art, such as deterioration in space efficiency, increase in weight, and increase in cost, can be solved.

According to one embodiment of the present invention, the battery module and the direct BMS are physically directly connected without a separate component such as a cable, so that durability against vibration can be improved compared to the conventional vehicle battery management systems, and accordingly, reliability of measured data can be increased even in the special measurement situations such as high-speed driving environments.

According to one embodiment of the present invention, any type of relay BMS that relays or collects information about a plurality of battery cells connected to each of the plurality of direct BMSs between the master BMS and the plurality of direct BMSs and transmits the information to the master BMS, may not be required.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically illustrates a configuration of a system for diagnosing a battery cell according to one embodiment of the present invention.

5

FIG. 2 schematically illustrates an internal configuration of a direct BMS according to one embodiment of the present invention.

FIG. 3 schematically illustrates configurations of a battery module and a battery cell connection unit according to one embodiment of the present invention.

FIG. 4 schematically illustrates an internal configuration of a first substrate unit according to one embodiment of the present invention.

FIG. 5 schematically illustrates an internal configuration of a second substrate unit according to one embodiment of the present invention.

FIG. 6 schematically illustrates an internal configuration of a signal transmission unit according to one embodiment of the present invention.

FIG. 7 schematically illustrates a process of transmitting driving power of the direct BMS according to one embodiment of the present invention.

FIG. 8 schematically illustrates an internal configuration for performing asynchronous wireless communication in each of the direct BMS and a master BMS according to one embodiment of the present invention.

FIG. 9 schematically illustrates a structure of a communication frame in the asynchronous wireless communication performed between the master BMS and a plurality of direct BMSs according to one embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 1 schematically illustrates a configuration of a system for diagnosing a battery cell according to one embodiment of the present invention.

Specifically, the system for diagnosing a battery cell of the present invention senses a plurality of information about a battery provided in a vehicle using electric power of the battery as power, and diagnoses the battery based on the sensed information. As illustrated in FIG. 1, the diagnosis system includes a plurality of direct BMSs **1000.1** to **1000.N** (hereinafter, **1000**) and a master BMS **2000**, and the master BMS **2000** wirelessly communicates with each of the plurality of direct BMSs **1000**. That is, in FIG. 1, a solid line represents portions that are physically connected or comes into contact with each other, and a dotted line represents configurations that are wirelessly connected to each other.

The diagnosis system may be electrically connected to a battery located inside the vehicle to measure a plurality of indices for managing the battery. The plurality of indices include a voltage, a current, and a temperature. Further, the battery, which will be described later, is connected to the diagnosis system of the present invention and has a configuration including a battery module including a plurality of battery cells and a battery pack including a plurality of battery modules according to a known technology, but the present invention is not limited thereto and may be applied through a simple design change as long as the battery includes a plurality of battery cells.

More specifically, each of the plurality of direct BMSs **1000** may be electrically connected to the plurality of battery cells to measure a voltage, a current, and a temperature of the plurality of connected battery cells. As illustrated in FIG. 1, in one embodiment of the present invention, each of a plurality of direct BMSs **1000** is electrically connected to one battery module. Meanwhile, in another embodiment of the present invention, the unit of the plurality of battery

6

cells connected to each of the plurality of direct BMSs **1000** is not limited to a battery module, and the unit may be set as a battery cell group including a predetermined number of battery cells.

Hereinafter, the diagnosis system will be described based on a configuration in which the plurality of direct BMSs **1000** are connected to the plurality of battery modules, respectively, and it is preferable that the number of the plurality of direct BMSs **1000** included in the diagnosis system of the present invention corresponds to the number of battery modules included in the battery.

Information sensed by each of the plurality of direct BMSs **1000** from the battery module connected thereto is transmitted to the master BMS **2000** through a first wireless communication unit **2200** (see FIG. 2) and a first antenna unit **2300** (see FIG. 2) included in each of the plurality of direct BMSs **1000**, and the master BMS **2000** diagnoses a state of the battery based on the information received from each of the plurality of direct BMSs **1000**. Hereinafter, a hardware configuration of the system for diagnosing a battery cell of the present invention and a configuration in which the direct BMS **1000** communicates with the master BMS **2000** will be described in detail.

FIG. 2 schematically illustrates an internal configuration of the direct BMS **1000** according to one embodiment of the present invention.

As illustrated in FIG. 2, a system for diagnosing a battery cell located inside a vehicle includes: a plurality of direct BMSs **1000** located inside the vehicle and electrically connected to a plurality of battery cells included in a battery module; and a master BMS **2000** located inside the vehicle and configured to wirelessly communicate with the direct BMSs **1000**; wherein the direct BMS **1000** includes: a first substrate unit **1000** operable in a first power range and including a substrate and an element; and a second substrate unit **2000** operable in a second power range lower than the first power range and including a substrate and an element, the first substrate unit **1000** includes: a battery cell connection unit **1100** including ports that physically and electrically come into direct contact with an exposed electrode of each of the plurality of battery cells; a voltage measurement unit **1200** electrically connected to the battery cell connection unit **1100** and configured in a form of a circuit to measure a voltage of each of the battery cells; and a current measurement unit **1300** electrically connected to the battery cell connection unit **1100** and configured in a form of a circuit to measure a current of the battery cell or the battery module, and the second substrate unit **2000** includes: an MCU **2100**; a first wireless communication unit **2200** connected to the MCU **2100** to wirelessly communicate with the master BMS **2000**; and a power supply unit **2400** configured to supply power to an element inside the second substrate unit **2000** by receiving power from the first substrate unit **1000** while lowering a voltage of power of the plurality of battery cells.

Briefly, FIG. 2 illustrates only some of the internal configurations of the direct BMS **1000**, and more specific internal configurations will be described later with reference to the descriptions of FIGS. 4 to 8. The direct BMS **1000** mainly includes the first substrate unit **1000** and the second substrate unit **2000** which are implemented by different processes. The first substrate unit **1000** and the second substrate unit **2000**, which are implemented by different processes, are configured as one system semiconductor chip by applying a multi chip package (MCP) technology.

In detail, the battery cell connection unit **1100** included in the first substrate unit **1000** includes a plurality of ports, and electrodes of each of the plurality of battery cells connected

to the corresponding direct BMS **10000** physically comes into direct contact with each other without a separate cable. Through the above configuration, it is possible to solve the following problems in the related art.

In the conventional battery management system technology, the plurality of battery cells are grouped, and signal output lines of the grouped battery cells are connected to a single slave BMS. For example, when it is assumed that 25 battery cells are present in one battery module, 5 of the 25 battery cells are grouped into 5 groups, and signals obtained by measuring voltages and currents of all of the 5 battery cells are transmitted to the slave BMS through separate wired communication. In addition, since signals transmitted from each of the 5 signal lines may not be simultaneously transmitted to the slave BMS, a module capable of switching channels between the 5 signal lines and the slave BMS is required.

Therefore, one slave BMS receives information through a total of 5 signal lines for 25 battery cells, and in this process, there is a problem that noise is generated by the switching module. Further, in the battery management system having the above configuration, an isolator for separating a signal component is required to be applied or installed at a plurality of positions in order to prevent electro-magnetic interference from being induced between a plurality of signal lines.

That is, in order to solve problems that may occur due to switching noise or a separating element in the related art, the present invention has a configuration of the battery cell connection unit **1100** in which each of the plurality of ports physically comes into direct contact with the electrodes of each of the plurality of battery cells. The configuration of the battery cell connection unit **1100** will be described later with reference to the description of FIG. 3.

The first substrate unit **1000** includes a voltage measurement unit **1200**, a current measurement unit **1300**, and a temperature measurement unit **1400**. The voltage measurement unit **1200** is configured in the form of a circuit and measures the voltage of each of the plurality of battery cells included in the battery module connected to the direct BMS **10000** through the battery cell connection unit **1100**. More specifically, referring to FIGS. 2 and 3, the battery cell connection unit **1100** includes a plurality of voltage measurement ports, in which the number of voltage measurement ports corresponds to the number of battery cells included in the battery module, and the voltage measurement unit **1200** may measure the voltage of each of the plurality of battery cells of the corresponding battery module through the voltage measurement port.

In this case, the voltage is measured based on a RAW voltage signal for each battery cell, and the RAW voltage signal means an electrical signal which is not interfered or distorted by an external factor. That is, the voltage measurement unit **1200** may derive voltage information precisely and accurately measured for each of the plurality of battery cells based on the RAW voltage signal of each of the plurality of battery cells received through the plurality of voltage measurement ports, and then may transmit the voltage information to the MCU **2100** of the second substrate unit **2000**, and the MCU **2100** may transmit the received voltage information to the master BMS **20000** through a wireless communication scheme.

Additionally, the voltage measurement unit **1200** may be connected to a high voltage alarm module (not illustrated) and a low voltage alarm module (not illustrated), and when the voltage of one or more specific battery cells is higher than a predetermined reference, the high voltage alarm module or the low voltage alarm module may immediately

generate an alarm message to transmit the alarm message to the master BMS **20000** through the first wireless communication unit **2200**. In one embodiment of the present invention, when the high voltage alarm message or the low voltage alarm message is transmitted to the master BMS **20000**, the master BMS **20000** may command the corresponding direct BMS **10000** to perform cell balancing so as to adjust the voltage between cells, or may cut off power of the corresponding direct BMS **10000**.

The current measurement unit **1300** is configured in the form of a circuit and measures the current of the battery module connected to the corresponding direct BMS **10000** through the current measurement port. More specifically, the current measurement port may be located at an output terminal of a battery module or a battery cell group connected to the direct BMS **10000**, and the current measurement unit **1300** is preferably provided to electrically and physically come into direct contact with the corresponding battery module.

The current measured through the current measurement port is measured based on a RAW current signal for the corresponding battery module, and the RAW current signal means an electrical signal which is not interfered or distorted by an external factor like the RAW voltage signal described above. That is, the current measurement unit **1300** measures accurate current information about the corresponding battery module based on the RAW current signal for the battery module received through the current measurement port, and then transmits the measured accurate current information to the MCU **2100**, and the MCU **2100** transmits the received current information to the master BMS **20000** in a wireless communication manner.

Additionally, the current measurement unit **1300** may be connected to an overcurrent alarm module (not illustrated), and when the current of the corresponding battery module is higher than a predetermined reference, the overcurrent alarm module may immediately generate an alarm message to transmit the alarm message to the master BMS **20000** through the first wireless communication unit **2200**.

Meanwhile, according to another embodiment of the present invention, the battery cell connection unit **1100** includes a plurality of current measurement ports, and the plurality of current measurement ports are connected to each of the battery cells to measure the current of each of the plurality of battery cells like the voltage measurement port illustrated in FIG. 4. However, in the present specification, a description will be made based on one embodiment in which the current measurement port is provided at the output terminal of the battery module or the last of the plurality of battery cells to measure the current of the battery module or all of the plurality of battery cells.

The temperature measurement unit **1400** is provided on the first substrate unit **1000** together with the voltage measurement unit **1200** and the current measurement unit **1300**, and includes a temperature sensor. The temperature measurement unit **1400** may measure a temperature of the first substrate unit **1000** through the temperature sensor. According to one embodiment of the present invention, the temperature measurement unit **1400** may be connected to an external temperature sensor, and the external temperature sensor may measure a temperature of the battery module connected to the direct BMS **10000** and transmit a measurement result to the temperature measurement unit **1400**.

The temperature measurement unit **1400** transmits the temperature result measured for the corresponding battery module to the MCU **2100**, and the MCU **2100** transmits the corresponding result to the master BMS **20000** through the

first wireless communication unit **2200**. The temperature measurement unit **1400** is connected to an overheating alarm module (not illustrated), and when the temperature measured by the temperature measurement unit **1400** is a higher than predetermined reference, the overheating alarm module may immediately generate an alarm message to transmit the alarm message to the master BMS **20000** through the first wireless communication unit **2200**. In addition, the temperature measurement unit **1400** may be connected to the voltage measurement unit **1200** to use some of the power introduced into the voltage measurement unit **1200**.

As described above, the first substrate unit **1000** includes a configuration to sense information about the battery module connected to the corresponding direct BMS **10000**. Meanwhile, as illustrated in FIG. 2, the first substrate unit **1000** may be designed based on individual cells, it is possible to achieve low-power operation compared to the related art in which voltages of all of the plurality of battery cells are measured.

In addition, the first substrate unit **1000** further includes configurations such as a regulator **1500** and a signal transmission unit **1600** in addition to the configurations described above, and a detailed description thereof will be described later with reference to the description of FIG. 4.

Meanwhile, the second substrate unit **2000** includes a configuration to receive information sensed by the first substrate unit **1000** to transmit the information to the master BMS **20000**. More specifically, the MCU **2100** of the second substrate unit **2000** may receive the results measured by the voltage measurement unit **1200**, the current measurement unit **1300**, and the temperature measurement unit **1400**, perform data processing, and then transmit the results to the first wireless communication unit **2200**. In addition, the MCU **2100** of the second substrate unit **2000** controls each of the detailed configurations of the direct BMS **10000** including the voltage measurement unit **1200**, the current measurement unit **1300**, and the temperature measurement unit **1400**.

The first wireless communication unit **2200** is connected to the MCU **2100** to wirelessly communicate with the master BMS **20000**. As in FIG. 5, the illustrated first wireless communication unit **2200** includes a modem unit **2210** and a transceiver unit **2220**, and performs steps of: modulating binarized data, which is received from the MCU **2100**, into a phase shift modulation signal (PSMS) and modulating the phase shift modulation signal into a differential phase shift modulation signal (DPSMS) to transmit the PSMS and DPSMS to the outside, in order to transmit the data received from the MCU **2100** to the master BMS **20000**; and demodulating the differential phase shift modulation signal (DPSMS), which is received from the first antenna unit **2300**, into a phase shift modulation signal (PSMS) and demodulating the decoded phase shift modulation signal (PSMS) into binarized data to transmit the binarized data to the MCU **2100**, in order to receive the data from the master BMS **20000**.

The first antenna unit **2300** connected to the first wireless communication unit **2200** is configured in the form of a circuit. The antenna unit **2300** receives a signal modulated by the first wireless communication unit **2200** to broadcast the signal to the outside, or receives the signal broadcasted by the master BMS **20000** to transmit the signal to the first wireless communication unit **2200**. Referring to FIG. 8, the signal broadcasted by the first antenna unit **2300** may be received by the second antenna unit **23000** of the master BMS **20000**. In a preferred embodiment, an antenna

included in the first antenna unit **2300** may transmit and receive radio waves in a 900 MHz band.

Meanwhile, as illustrated in FIG. 2, the first substrate unit **1000** and the second substrate unit **2000** are illustrated as being physically provided at different positions, but in the direct BMS **10000** of the present invention, the MCP process may be applied so that the first substrate unit **1000** and the second substrate unit **2000** may be provided on the same semiconductor chip. In more detail, the direct BMS **10000** may be implemented in such a manner that the second substrate unit **2000** is stacked on the first substrate unit **1000** or the first substrate unit **1000** is stacked on the second substrate unit **2000**.

The MCP process is a process technology in which two or more semiconductor chips are stacked to form one package, and the present invention uses the MCP process to implement a more miniaturized direct BMS **10000** and to reduce manufacturing costs. Additionally, according to one embodiment of the present invention, a flash memory **3000** illustrated in FIG. 2 may also be implemented by being stacked on the same semiconductor chip as the first substrate unit **1000** and the second substrate unit **2000**.

FIG. 3 schematically illustrates a configuration of the battery module and the battery cell connection unit **1100** according to one embodiment of the present invention.

As illustrated in FIG. 3, the positive electrode terminal and the negative electrode terminal of each of the plurality of battery cells comes into direct contact with ports provided in the battery cell connection unit **1100**, and the voltage measurement unit **1200** is connected to each of the ports to measure a voltage of each of the battery cells.

Specifically, as described above, the voltage measurement unit **1200** of the direct BMS **10000** may receive a RAW voltage signal for each of the plurality of battery cells, which is included in the battery module connected to the direct BMS **10000**, through the plurality of voltage measurement ports included in the battery cell connection unit **1100** to measure the accurate voltage of each battery cell.

More specifically, the plurality of battery cells in the battery module are connected in series, and in a preferred embodiment, the plurality of battery cells in the battery module may be disposed such that a (+) electrode and a (-) electrode intersect each other. The exposed (+) electrode and the exposed (-) electrode of each of the battery cells physically and electrically come into direct contact with a (+) contact portion and a (-) contact portion of the battery cell connection unit **1100**, respectively. That is, the (+) electrode of the battery cell corresponds to a configuration that is directly connected to the (+) contact portion located inside the battery cell connection unit **1100**, and the (-) electrode of to a configuration that is the battery cell corresponds directly connected to the (-) contact portion located inside the battery cell connection unit **1100**.

In addition, referring to FIG. 3, each of the plurality of voltage measurement ports connected to the voltage measurement unit **1200** physically and electrically comes into direct contact with one (+) contact portion and one (-) contact portion at the same time. Through the above configuration, each of the plurality of voltage measurement ports may receive a RAW voltage signal for each of the battery cells, and the voltage measurement unit **1200** may accurately measure a voltage for the corresponding battery cell based on the received RAW voltage signal.

That is, unlike the conventional battery management system technology, the configuration for measuring the voltage in the present invention does not cause interference or loss in the process in which data or a signal, starting from

the battery cell, is transmitted to the direct BMS **10000**, and thus the RAW voltage signal may be directly transmitted to the direct BMS **10000**, and the voltage measurement unit **1200** in the direct BMS **10000** may measure the accurate voltage based on the transmitted RAW voltage signal. Therefore, the above-described configuration may exclude an EMI isolator included in the conventional battery management system.

Meanwhile, as illustrated in FIG. 4, the current measurement port of the battery cell connection unit **1100** also physically and electrically comes into direct contact with the battery module, so that the RAW current signal may be directly transmitted to the direct BMS **10000**, and the current measurement unit **1300** in the direct BMS **10000** may accurately and precisely measure the current flowing in the battery module based on the RAW current signal.

In addition, in one embodiment of the present invention, the voltage measurement unit **1200** and the current measurement unit **1300** may measure the voltage and the current of the battery cell over time, and may transmit waveforms of the output voltage and waveforms of the output current of the corresponding battery cell over time to the MCU **2100**.

Additionally, individual impedance of each of the plurality of battery cells may be derived through the configuration of the direct BMS **10000** that is directly connected to the battery module, and the individual impedance of each of the plurality of battery cells is derived so that abnormal battery cells in the battery module may be individually detected. For example, it is possible to prevent problems such as fire, which may be caused due to non-balance of voltages between the plurality of battery cells, in an early stage, and this corresponds to an improved configuration and effect compared to the related art in which the average or total voltage in the battery module is measured.

FIG. 4 schematically illustrates an internal configuration of the first substrate unit **1000** according to one embodiment of the present invention.

As illustrated in FIG. 4, the first substrate unit **1000** includes a signal transmission unit **1600** configured to receive measurement information including the current and the voltage of the plurality of battery cells measured by the current measurement unit **1300** and the voltage measurement unit **1200**, respectively, to output the measurement information to the second substrate unit **2000** as an analog electrical signal, and the first substrate unit **1000** is manufactured by the LDCMOS process and operates in the first power range of 32 V to 36 V.

In addition, the first substrate unit **1000** includes the regulator **1500** configured to supply power to the power supply unit **2400** by receiving the power of all of the plurality of battery cells.

In detail, the first substrate unit **1000** of the direct BMS **10000** further includes the regulator **1500** and the signal transmission unit **1600**, in addition to the above-described configurations of the battery cell connection unit **1100**, the voltage measurement unit **1200**, the current measurement unit **1300**, and the temperature measurement unit **1400**. The regulator **1500** transmits the power to the second substrate unit **2000** by constantly reducing the power transmitted through the battery cell connection unit **1100**, and the power transmitted to the second substrate unit **2000** is used as driving power of the MCU **2100**, first wireless communication unit **2200**, and first antenna unit **2300** of the second substrate unit **2000**. In addition, the power output from the regulator **1500** may be used as driving power of a configuration included in the first substrate unit **1000** or a signal required to operate a specific configuration.

Meanwhile, in one embodiment of the present invention, the first substrate unit **1000** may further include a configuration for reducing the voltage received from the battery module in addition to the regulator **1500**, and through the above configuration, the first substrate unit **1000** may be implemented by the LDCMOS process in which the first substrate unit **1000** operates in a power range that is lower than that of the conventional BDC process.

More specifically, the conventional battery management system uses a chip, a module, or the like that is implemented by the BCD process. The BCD process refers to a process technology for integrating three processes, including Bipolar, CMOS, and DMOS, into one process. The BCD process has an advantage of withstanding high voltage, and thus has been used in power semiconductors, motor drive ICs, LED driver ICs, and the like. In other words, in the conventional battery management system, a high voltage is required to diagnose the battery, and in order to withstand the high voltage, a chip or a module implemented by the BCD process, which is capable of withstanding the voltage up to several hundreds of volts, is used. Meanwhile, it is impossible for the battery management system using the BCD process to sense or diagnose precise information about the battery due to process reasons.

In order to overcome the above-described disadvantages of the BCD process, the inventor of the present invention adds a configuration for dropping the high voltage outputted from the plurality of battery cells to the first substrate unit **1000**, thereby lowering the voltage outputted from the plurality of battery cells to the first power range of 32 V to 36 V. That is, by using a chip or a module implemented by the LDCMOS process corresponding to a low-power process instead of the BCD process which makes it difficult to precisely measure the battery due to process characteristics, it is possible to more accurately measure the voltage and the current for a battery module connected to the corresponding direct BMS **10000** compared to the related art. In this case, the RAW voltage signal measured by the voltage measurement unit **1200** corresponds to a signal before passing through the configuration for dropping the voltage described above.

Meanwhile, as illustrated in FIG. 4, the regulator **1500** is simultaneously connected to the battery cell connection unit **1100** and the temperature measurement unit **1400**. In more detail, the regulator **1500** may supply the power to the second substrate unit **2000** by receiving the power input from the battery cell connection unit **1100** and the signal received from the temperature measurement unit **1400**, and more specifically, the power may be supplied to the power supply unit **2400** of the second substrate unit **2000** with reference to FIG. 2. The regulator **1500** will be described later with reference to the description of FIG. 7.

The signal transmission unit **1600** transmits the measurement information generated by the first substrate unit **1000** to the second substrate unit **2000**. More specifically, as illustrated in FIG. 4, the signal transmission unit **1600** transmits measurement information including the analog voltage signal generated by the voltage measurement unit **1200** and the analog current signal generated by the current measurement unit **1300** to the MCU **2100** of the second substrate unit **2000**. In one embodiment of the present invention, the measurement information may further include temperature information generated by the temperature measurement unit **1400**. The signal transmission unit **1600** will be described later in more detail with reference to the description of FIG. 6.

13

As described above, the direct BMS **1000** of the present invention separately includes a configuration for transmitting the power and a configuration for transmitting the information between the first substrate unit **1000** and the second substrate unit **2000**, so that it is possible to achieve low-power operation and more accurately diagnose a state of the battery.

FIG. **5** schematically illustrates an internal configuration of the second substrate unit **2000** according to one embodiment of the present invention.

As illustrated in FIG. **5**, the second substrate unit **2000** operates in the second power range of 5 V or less and includes an ADC **2500**, in which the ADC **2500** converts the analog electrical signal received from the signal transmission unit **1600** into a digital electrical signal to transmit the digital electrical signal to the MCU **2100**.

In addition, the second substrate unit **2000** further includes: a modem unit **2210** including a modem transmission unit **2211** configured to perform modulation for transmitting the information received from the MCU **2100** to the master BMS **20000**, and a modem reception unit **2212** configured to perform demodulation for transmitting the signal received from the master BMS **20000** to the MCU **2100**; and a first wireless communication unit **2200** including a transceiver unit **2220** that includes: a Tx unit **2221** configured to broadcast the signal modulated by the modem transmission unit **2211** through the first antenna unit **2300**; and an Rx unit **2222** configured to process the signal received through the first antenna unit **2300**.

Specifically, referring to the description of FIG. **4**, the signal transmission unit **1600** transmits measurement information including the analog voltage signal and the analog current signal, and the analog to digital converter (ADC) **2500** of the second substrate unit **2000** receives the measurement information and converts the measurement information into a digital signal. Meanwhile, the second substrate unit **2000** operates in a second power range (5 V or less) less than the first power range (32 V to 36 V) in which the first substrate unit **1000** implemented by the LDCMOS process operates.

That is, the second substrate unit **2000** and the first substrate unit **1000** are implemented by different processes, and the ADC **2500** is included in the second substrate unit **2000** driven with less power as described above.

More specifically, the ADC **2500** for converting an analog signal into a digital signal has characteristics in that a size thereof is increased in a process for a high voltage due to process characteristics, and a processing speed thereof is decreased so that performance thereof is deteriorated. That is, when the ADC **2500** is provided on a chip designed for a process for a low voltage, high performance may be exhibited through the ADC **2500** having a smaller size. However, as described above, in the conventional battery management system technology, a chip or a module implemented by the BCD process, which is a process for high voltage, has been used for durability, and accordingly, an ADC having a large size and slightly poor performance has to be used.

However, in the present invention, in order to solve the problem, a configuration (the regulator **1500**, the signal transmission unit **1600**, etc.) for dropping a voltage is added to the first substrate unit **1000**, and the ADC **2500** is provided on the second substrate unit **2000** implemented by a process for a low voltage of 5 V or less, thereby reducing the size of the ADC **2500** and improving performance.

The ADC **2500** of the present invention, which has the above-described technical features, converts the analog elec-

14

trical signal received from the signal transmission unit **1600** into a digital electrical signal to transmit the electrical signal to the MCU **2100**. Thereafter, the MCU **2100** performs signal processing to transmit data based on the information, which is received from the ADC **2500**, to the master BMS **20000** through the first wireless communication unit **2200** and the first antenna unit **2300** through a wireless communication scheme.

As illustrated in FIG. **5**, the first wireless communication unit **2200** includes the modem unit **2210** and the transceiver unit **2220**, in which the modem unit **2210** includes the modem transmission unit **2211** configured to perform modulation for transmitting the information received from the MCU **2100** to the master BMS **20000**, and the modem reception unit **2212** configured to perform demodulation for transmitting the signal received from the master BMS **20000** to the MCU **2100**.

In addition, the transceiver unit **2220** includes the Tx unit **2221** including a configuration to broadcast the signal modulated by the modem transmission unit **2211** through the first antenna unit **2300**, and the Rx unit **2222** including a configuration to process the signal received through the first antenna unit **2300**.

More specifically, the modem transmission unit **2211** includes a hardware and/or software configuration to modulate digital data received from the MCU **2100** into a phase shift modulation signal (PSMS) and modulate the phase shift modulation signal into a differential phase shift modulation signal (DPSMS), and the modem reception unit **2212** includes a hardware and/or software configuration to demodulate the received differential phase shift modulation signal (DPSMS) into a phase shift modulation signal (PSMS) and demodulate the demodulated phase shift modulation signal (PSMS) into a digital signal.

Meanwhile, the Tx unit **2221** includes a hardware and/or software filter for processing the signal modulated by the modem transmission unit **2211**, a digital to analog converter (DAC), and the like, and the Rx unit **2222** includes a hardware and/or software filter for processing the signal received by the first antenna unit **2300**, the ADC **2500**, and the like.

That is, the information received from the MCU **2100** of the direct BMS **10000** may be transmitted to the first antenna unit **2300** via the modem transmission unit **2211** and the Tx unit **2221** and transmitted to the master BMS **20000**, and the information transmitted from a control unit **21000** of the master BMS **20000** is received through the first antenna unit **2300** and transmitted to the MCU **2100** via the Rx unit **2222** and the modem reception unit **2212**.

As described above, through the wireless communication scheme using the differential phase shift modulation signal (DPSMS) and the phase shift modulation signal (PSMS), it is possible to achieve more stable wireless communication in an environment in which line of sight (LOS) is not secured inside the battery pack of the vehicle due to aluminum shielding or the like.

In addition, the wireless communication scheme of the present invention does not include a configuration related to a communication station that relays the communication, which is adopted in a conventional battery management system including a slave BMS or a module BMS, thereby achieving a more simplified configuration of a wireless communication system.

In addition, the wireless communication scheme of the present invention performs asynchronous wireless communication, thereby not requiring a separate configuration for synchronization during wireless communication, and it is

15

possible to solve problems such as deterioration of space efficiency, increase in weight, increase in cost, and the like, which occur when an isolator for preventing electromagnetic interference from being induced in a conventional wired BMS-related technology is applied or installed at a plurality of locations.

FIG. 6 schematically illustrates an internal configuration of the signal transmission 1600 according to one embodiment of the present invention.

As illustrated in FIG. 6, the signal transmission unit 1600 includes: a multiplexer (MUX) unit 1610 including a circuit configured to output one of electrical signals for each of the plurality of battery cells under a control of the MCU 2100; a voltage dropping unit 1620 including a circuit configured to drop a voltage signal received from the voltage measurement unit 1200 under the control of the MCU 2100; and a filter unit 1630 including a circuit configured to reduce noise of the electrical signal received from the voltage measurement unit 1200, and the signal transmission unit 1600 transmits an analog voltage signal to the ADC 2500 after reducing a magnitude of the voltage signal, which is transmitted through the battery cell connection unit, to a predetermined reference or less.

FIG. 6 schematically illustrates a configuration in which the analog voltage signal of each of the plurality of battery cells is generated by the first substrate unit 1000, and then transmitted to the second substrate unit 2000.

In detail, referring to FIGS. 3 and 4, the voltage of each of the plurality of battery cells is measured by the voltage measurement unit 1200 through the plurality of voltage measurement ports included in the battery cell connection unit 1100. In this case, the analog voltage signal generated by the voltage measurement unit 1200 is transmitted to the MUX unit as illustrated in FIG. 6.

The MUX unit 1610 includes a circuit for outputting one of the electrical signals for each of the plurality of battery cells. For example, in a case of the direct BMS 10000 to which the battery module including battery cells #1 to #8 is connected, the RAW voltage signal of each of battery cells #1 to #8 is transmitted to the voltage measurement unit 1200 through the battery cell connection unit 1100, and the analog voltage signals (for example, voltage signals #1 to #8) for battery cells #1 to #8 generated by the voltage measurement unit 1200 are transmitted to the MUX unit 1610, respectively. Thereafter, the MUX unit 1610 transmits one of voltage signals #1 to #8 to the voltage dropping unit 1620 according to a predetermined rule.

According to one embodiment of the present invention, the MUX unit 1610 may output one of the plurality of electrical signals to the voltage dropping unit 1620 under the control of the MCU 2100 of the second substrate unit 2000, according to another embodiment of the present invention, a configuration may be added between the battery cell connection unit 1100 and the MUX unit 1610 to amplify the voltage signal or the current signal transmitted from the plurality of battery cells, to convert impedance so as to be suitable for connection with a circuit at a rear end, or to protect the circuit itself from an external noise/static electricity, and according to still another embodiment of the present invention, a configuration including a circuit that is adjusted to be suitable for connection with the circuit at the rear end by lowering or raising a level of the voltage signal received from the plurality of battery cells may be further added.

The voltage drop unit 1620 includes a circuit for dropping the voltage signal received from the MUX unit 1610 under the control of the MCU 2100. In a preferred embodiment,

16

the voltage level of the voltage signal received from the MUX unit 1610 may be lowered to $\frac{1}{2}$ or $\frac{1}{4}$, and the second substrate unit 2000 may be driven in a power range that is smaller than that of the first substrate unit 1000 through the configuration for adjusting the level of the voltage signal described above or the configuration of the voltage dropping unit 1620.

The analog voltage signal passing through the voltage dropping unit 1620 is transmitted to a first substrate connection unit 1640 via the filter unit 1630. The filter unit 1630 includes a circuit for reducing unnecessary high frequency noise included in the electrical signal received from the voltage measurement unit 1200, and may include a low pass filter (LPF) in a preferred embodiment.

The analog voltage signal passing through the filter unit 1630 may be transmitted to the second substrate unit 2000 through the first substrate connection unit 1640, and more specifically, may be transferred to the ADC 2500 of the second substrate unit 2000, converted into a digital signal, and then transferred to the MCU 2100. Although FIG. 6 illustrates only a configuration in which the first substrate connection unit 1640 receives only the analog voltage signal, referring to FIG. 4, the first substrate connection unit 1640 may receive a current signal and temperature information from the current measurement unit 1300 and the temperature measurement unit 1400 to transmit the received current signal and temperature information to the ADC 2500.

FIG. 7 schematically illustrates a process of transmitting driving power of the direct BMS 10000 according to one embodiment of the present invention.

As illustrated in FIG. 7, the regulator 1500 supplies the power to the power supply unit 2400 by lowering the magnitude of the voltage signal of all of the plurality of battery cells to a predetermined reference or less when a reference temperature is received and the temperature information received from the temperature measurement unit 1400 is less than a predetermined reference, and stops power supply to the power supply unit 2400 when the reference signal is not received or the temperature information received from the temperature measurement unit 1400 is equal to or greater than the predetermined reference.

In detail, the regulator 1500 receives the power of all of the plurality of battery cells through the battery cell connection unit 1100, lowers the received power, and then transmits the lowered power to the power supply unit 2400 of the second substrate unit 2000. As illustrated in FIG. 7, the regulator 1500 receives the entire power from the battery cell connection unit 1100, and according to one embodiment of the present invention, the regulator 1500 may receive the power with a lowered voltage of a predetermined reference or less through a configuration including a circuit for appropriately adjusting the level of the voltage signals received from the plurality of battery cells to be suitable for connection with the circuit of the rear end by appropriately lowering or raising the level of the voltage signals. The reference signal is a signal including a high signal and a low signal, the regulator 1500 is turned on when the reference signal is input to the regulator 1500 as the high signal, and the regulator 1500 is turned off when the reference signal is input to the regulator 1500 as the low signal.

Meanwhile, the regulator 1500 may be turned on/off according to the temperature information received from the temperature measurement unit 1400 as well as the reference signal. More specifically, when the temperature information transmitted from the temperature measurement unit 1400 is equal to or higher than a preset temperature, the regulator

1500 is turned off even if the high signal is input as the reference signal. In other words, the regulator 1500 operates even when the high reference signal is input to the regulator 1500 and the temperature information of equal to or lower than a predetermined temperature is received from the temperature measurement unit 1400, and accordingly, the driving power is cut off at a specific temperature or higher, thereby protecting the BMS, the battery, and an electric vehicle equipped with the battery.

In addition, in one embodiment of the present invention, the regulator 1500 may determine whether or not the temperature information transmitted from the temperature measurement unit 1400 is equal to or higher than a predetermined reference temperature, but in another embodiment of the present invention, the overheating alarm module connected to the temperature measurement unit 1400 may determine whether or not a temperature of a substrate is equal to or higher than a predetermined reference and may transmit determination information to the regulator 1500.

As described above, when the reference temperature is equal to or lower than the predetermined reference temperature and the high reference signal is input to the regulator 1500, the regulator 1500 transmits the power to the power supply unit 2400 of the second substrate unit 2000 by lowering the power received through the battery cell connection unit 1100 to a predetermined level. The power supply unit 2400 distributes the power to each of the detailed configurations included in the second substrate unit 2000, and in this case, it is preferable that the distributed power corresponds to the second power range (5 V or less) described above.

Although FIG. 7 illustrates a configuration in which the power supply unit 2400 is connected to only the MCU 2100, the first wireless communication unit 2200, and the first antenna unit 2300, the power supply unit 2400 may be connected to the respective configurations (not illustrated in FIG. 2) to supply the power to the corresponding configurations.

FIG. 8 schematically illustrates an internal configuration for performing asynchronous wireless communication in each of the direct BMS 10000 and the master BMS 20000 according to one embodiment of the present invention.

As illustrated in FIG. 8, each of the plurality of direct BMSs 10000 includes the first wireless communication unit 2200 and the first antenna unit 2300 configured to wirelessly communicate with the master BMS 20000, and the master BMS 20000 includes the second wireless communication unit 22000 and the second antenna unit 23000 configured to wirelessly communicate with each of the plurality of direct BMSs 10000. The first wireless communication unit 2200 is connected to the MCU 2100 to receive the information generated by the first substrate unit 1000, and the second wireless communication unit 22000 is connected to the control unit 21000 to transmit the data received from the plurality of direct BMSs 10000 to the control unit 21000.

The control unit 21000 may include one or more memories and one or more processes, and may diagnose a state of each of the battery cells or a state of all of the battery modules connected to the plurality of direct BMSs 10000 based on the data received through the second wireless communication unit 22000. In one embodiment of the present invention, the control unit 21000 may determine the state of the entire battery provided in the corresponding vehicle based on a diagnosis result for the state of the battery module or the battery cell. For example, the control unit 21000 may derive the possibility of fire that may occur during charging, or may derive the remaining traveling

distance in real time in consideration of the current traveling state and the battery state during traveling.

Additionally, the control unit 21000 may broadcast control information about each of the plurality of direct BMSs 10000 through the second wireless communication unit 22000 and the second antenna unit 23000. The wireless communication between the master BMS 20000 and the plurality of direct BMSs 10000 will be described later with reference to the description of FIG. 9.

FIG. 9 schematically illustrates a structure of a communication frame in the asynchronous wireless communication performed between the master BMS 20000 and the plurality of direct BMSs 10000 according to one embodiment of the present invention.

As illustrated in FIG. 9, a communication frame of a wireless communication channel, through which the plurality of direct BMSs transmit and receive data to and from the master BMS 20000, includes a sub-frame including a downlink frame broadcasted by the master BMS 20000 to the plurality of direct BMSs 10000; and a plurality of uplink frames transmitted by each of the plurality of direct BMSs 10000 to the master BMS 20000, in which the downlink frame and the plurality of uplink frames in the sub-frame are temporally divided.

In addition, the data received by the plurality of direct BMSs 10000 through the downlink frame includes: identifier information of each of the direct BMSs 10000; and time division information of an uplink frame related to the direct BMS 10000 having the identifier information, and each of the direct BMSs 10000 transmits the data to the master BMS 20000 through the uplink frame allocated thereto, based on the time division information about its own the uplink frame included in the data received from the downlink frame.

Further, the wireless communication channel has a plurality of uplink frames that are temporarily configured following the downlink frame according to a predetermined rule, the data broadcasted in the downlink frame includes request information about a target to be transmitted by each of the direct BMSs 10000, and each of the direct BMSs 10000 generates response information according to the request information included in the data broadcasted in the downlink frame to transmit the data including the response information to the master BMS 20000 according to the time division information of its own uplink frame.

Further, the communication frame of the wireless communication channel includes a plurality of identical sub-frames, and the master BMS 20000 finally determines data to be received based on the received plurality of identical uplink frames.

Specifically, in one embodiment of the present information, a communication frame (=1 frame), which corresponds to a wireless communication period between the direct BMS 10000 and the master BMS 20000, may be set to a length of 100 ms. In a typical vehicle, a data transmission/reception period corresponds to 100 ms, and a response time of several sensor nodes connected to the direct BMS 10000 has a period of 100 ms, and thus it is most preferable to set 100 ms as a length of the communication frame, and accordingly, a state of each of the plurality of battery cells may be determined most quickly.

Meanwhile, as illustrated in FIG. 9, by performing repeated communication four times during one communication frame, it is possible to respond to a communication error and enhance communication stability, and the period of the repeated communication is defined as a sub-frame. The sub-frame includes a downlink frame to which communication (downlink communication) from the master BMS

20000 to the direct BMS **10000** is allocated and an uplink frame to which communication (uplink communication) from the direct BMS **10000** to the master BMS **20000** is allocated.

The downlink frame corresponds to a time allocated for the master BMS **20000** to broadcast to the plurality of direct BMSs **10000**, and the signal broadcasted by the master BMS **20000** in the downlink frame includes identifier information of each of the plurality of direct BMSs **10000** and time division information of the uplink frame related to the direct BMS **10000** having the corresponding identifier information.

For example, the data, which is broadcasted by the master BMS **20000** in the downlink frame, includes information including: a first time according to identifier information of the direct BMS **#1 10000.1** and identifier information of the direct BMS **#1 10000.1**; and a second time according to identifier information of the direct BMS **#2 10000.2** and identifier information of the direct BMS **#2 10000.2**. The direct BMS **#1 10000.1**, which receives the data broadcasted by the master BMS **20000**, transmits data about a battery module connected to itself (the direct BMS **#1 10000.1**) at the first time, and the direct BMS **#2 10000.2** transmits data about a battery module connected to itself (the direct BMS **#2 10000.2**) at the second time. Meanwhile, the data broadcasted by the master BMS **20000** may further include identifier information of each of the direct BMSs **10000** and request information, command information, or control information according to the identifier information, and the direct BMS **10000** receiving the data may generate response information corresponding to the received request information, command information, or control information to transmit the response information to the master BMS **20000**.

Meanwhile, the number of sub-frames included in one communication frame, a period of a sub-frame, a period of an uplink frame, a period of a downlink frame, and the like illustrated in FIG. 9 correspond to one embodiment of the present invention, and correspond to a matter that is changeable at any time according to the intention of the designer.

Additionally, a communication frequency channel for communication between the direct BMS **10000** and the master BMS **20000** includes a plurality of narrow band channels divided at intervals of 200 KHz in the 900 MHz band. Referring to the description of FIG. 5, when the wireless communication is performed by using the modulation/demodulation scheme of the present invention, the wireless communication system of the present invention has a simple structure and uses a narrow bandwidth compared to a wireless communication system using other modulation/demodulation schemes, and thus, the wireless communication system may perform optimized wireless communication in a harsh environment with a limited space and a limited weight, like a battery for a vehicle. In addition, in one embodiment of the present invention, the wireless communication system of the present invention may use several tens of 200 kHz narrow band channels due to good frequency efficiency, and thus may implement one-to-many communication such as communication of the master BMS **20000**—the direct BMS **10000** in a small bandwidth.

In one embodiment of the present invention, as illustrated in FIG. 9, when a total of 25 direct BMSs **10000** are present, one sub-frame includes one downlink frame and 25 uplink frames. In this case, it is preferable that one uplink frame is allocated to each of the plurality of direct BMSs **10000** and is allocated to different time slots. That is, information about a time slot to which each of the plurality of uplink frames is allocated is defined as time division information.

In addition, when a total of four repeated communications are performed for one communication frame, one communication frame (=100 ms) includes four sub-frames (=25 ms*4=100 ms), and one sub-frame (=25 ms) includes one downlink frame (=5 ms) and a total of 25 uplink frames (=0.8 ms*25=20 ms).

In brief, the wireless communication system employed in the present invention corresponds to a system in which when the master BMS **20000** simultaneously transmits a message to the plurality of direct BMSs **10000**, each of the plurality of direct BMSs **10000** receiving the corresponding message transmits data to the master BMS **20000** according to an uplink frame allocated thereto. A wireless communication system used in a conventional battery diagnosis technology includes many uplink communications, and thus a communication frame is configured based on the uplink frame, but the present invention adopts a structure in which the downlink frame is placed at the forefront of the communication frame, thereby securing reliability of data transmission and reception at an early stage and more efficiently utilizing the communication band.

In addition, the wireless communication system employed in the present invention is a system optimized for the size of data required for real-time transmission and reception of battery information and the number of nodes in a vehicle using electric power of the battery as power, so that it is possible to quickly and accurately provide a user with information about the battery of the vehicle even when the vehicle is stopped or is being charged or traveled.

Additionally, according to one embodiment of the present invention, by implementing the above-described communication system of the present invention in the descriptions of FIGS. 5, 8, and 9, any type of relay BMS that relays or collects information about a plurality of battery cells connected to each of the plurality of direct BMSs between the master BMS and the plurality of direct BMSs and transmits the information to the master BMS, may not be required.

According to one embodiment of the present invention, the system is designed based on individual cells and power to be supplied when a state of the battery is sensed, and the system may thus be operated at a relatively low voltage, thereby using a substrate implemented by the LDCMOS process rather than the conventional BCD process, and accordingly, it is possible to improve sensing accuracy for a battery.

According to one embodiment of the present invention, a configuration is adopted in which the first substrate unit which may measure the voltage, current, and temperature for the plurality of battery cells and is implemented by the LDCMOS process, and the second substrate unit which may wirelessly transmit information measured by the first substrate unit to the master BMS are implemented by different processes, and the first substrate unit and the second substrate unit are combined into one system semiconductor through an MCP method, so that it is possible to reduce manufacturing costs.

According to one embodiment of the present invention, the ADC is included in the second substrate unit, not the first substrate unit implemented by the LDCMOS process, so that it is possible to maximize performance thereof while minimizing the size of the ADC.

According to one embodiment of the present invention, the direct BMS is configured to physically and electrically come into direct contact with the battery cells, thereby measuring the voltage and the current for the battery cells, so that noise generated due to channel switching can be prevented in the related art in which the battery and the

21

direct BMS are indirectly connected through a cable or the like, and accordingly, the reliability of data measured for the battery cells can be increased.

According to one embodiment of the present invention, a master BMS and a plurality of direct BMSs connected to each of a plurality of battery modules perform asynchronous wireless communication, so that the problems caused by applying or installing an isolator to a plurality of positions to prevent electro-magnetic interference from being induced in the wired communication system of the related art, such as deterioration in space efficiency, increase in weight, and increase in cost, can be solved.

According to one embodiment of the present invention, the battery module and the direct BMS are physically directly connected without a separate component such as a cable, so that durability against vibration can be improved compared to the conventional vehicle battery management systems, and accordingly, reliability of measured data can be increased even in the special measurement situations such as high-speed driving environments.

According to one embodiment of the present invention, any type of relay BMS that relays or collects information about a plurality of battery cells connected to each of the plurality of direct BMSs between the master BMS and the plurality of direct BMSs and transmits the information to the master BMS, may not be required.

Although the above embodiments have been described with reference to the limited embodiments and drawings, however, it will be understood by those skilled in the art that various changes and modifications may be made from the above-mentioned description. For example, even though the described descriptions may be performed in an order different from the described manner, and/or the described components such as system, structure, device, and circuit may be coupled or combined in a form different from the described manner, or replaced or substituted by other components or equivalents, appropriate results may be achieved.

Therefore, other implementations, other embodiments, and equivalents to the claims are also within the scope of the following claims.

What is claimed is:

1. A system for diagnosing a battery cell located inside a vehicle, the system comprising:

a plurality of direct BMSs located inside the vehicle and electrically connected to a plurality of battery cells included in a battery module; and

a master BMS located inside the vehicle and configured to wirelessly communicate with the direct BMSs,

wherein the direct BMS includes:

a first substrate unit operable in a first power range and including a substrate and an element; and

a second substrate unit operable in a second power range lower than the first power range and including a substrate and an element,

the first substrate unit includes:

a battery cell connection unit including ports that physically and electrically come into direct contact with an exposed electrode of each of the plurality of battery cells;

a voltage measurement unit electrically connected to the battery cell connection unit and configured in a form of a circuit to measure a voltage of each of the battery cells; and

a current measurement unit electrically connected to the battery cell connection unit and configured in a form of a circuit to measure a current of the battery cell or the battery module, and

22

the second substrate unit includes:

an MCU;

a first wireless communication unit connected to the MCU to wirelessly communicate with the master BMS; and

a power supply unit configured to supply power to an element inside the second substrate unit by receiving power from the first substrate unit, wherein the first substrate unit receives power of the plurality of battery cells, lowers the received power of the plurality of battery cells by lowering a magnitude of a voltage signal related to the power of the plurality of battery cells, and supplies the lowered power to the power supply unit.

2. The system of claim 1, wherein both a positive electrode terminal and a negative electrode terminal of each of the plurality of battery cells come into direct contact with the ports included in the battery cell connection unit, and

the voltage measurement unit is connected to each port to measure a voltage for each of the battery cells.

3. The system of claim 1, wherein the first substrate unit includes a signal transmission unit configured to receive measurement information including the current and the voltage of the plurality of battery cells measured by the current measurement unit and the voltage measurement unit, respectively, to output the measurement information to the second substrate unit as an analog electrical signal,

the second substrate unit includes an ADC, and

the ADC converts the analog electrical signal received from the signal transmission unit into a digital electrical signal to transmit the digital electrical signal to the MCU.

4. The system of claim 3, wherein the signal transmission unit includes:

a multiplexer (MUX) unit including a circuit configured to output one of electrical signals for each of the plurality of battery cells under a control of the MCU;

a voltage dropping unit including a circuit configured to drop a voltage signal received from the voltage measurement unit under the control of the MCU; and

a filter unit including a circuit configured to reduce noise of the electrical signal received from the voltage measurement unit, and

the signal transmission unit transmits an analog voltage signal to the ADC after reducing a magnitude of the voltage signal, which is transmitted through the battery cell connection unit, to a predetermined reference or less.

5. The system of claim 1, wherein the first substrate unit is manufactured by an LDCMOS process.

6. The system of claim 1, wherein the first substrate unit operates in the first power range of 32 V to 36 V, and the second substrate unit operates in the second range of 5 V or less.

7. The system of claim 1, wherein the first substrate unit includes a regulator configured to supply power to the power supply unit by receiving the power of all of the plurality of battery cells, and

the regulator supplies the power to the power supply unit by reducing a magnitude of a voltage signal of all of the plurality of battery cells to a predetermined reference or less when a reference signal is received and temperature information received from a temperature measurement unit is lower than a predetermined reference, and stops power supply to the power supply unit when the reference signal is not received or the temperature

information received from the temperature measurement unit is equal to or higher than the predetermined reference.

8. The system of claim 1, wherein the second substrate unit includes:

- a modem unit including a modem transmission unit configured to perform modulation for transmitting information received from the MCU to the master BMS, and a modem reception unit configured to perform demodulation for transmitting a signal received from the master BMS to the MCU; and
- a first wireless communication unit including a transceiver unit that includes: a Tx unit including a configuration to broadcast the signal modulated by the modem transmission unit through a first antenna unit; and an Rx unit including a configuration to process the signal received through the first antenna unit.

9. The system of claim 8, wherein a communication frequency channel for communication between the direct BMS and the master BMS includes a plurality of narrow band channels divided at intervals of 200 KHz in a 900 MHz band.

* * * * *